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Paradigms and Evidence

Week 6 Wednesday — What counts as convincing evidence?

NRES 315: Human Dimensions of Fish and Wildlife Management

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The method is never the whole story

On Monday, we used disciplines to see different parts of a management problem.

Today we ask the prior question: **What assumptions lead a team to call one kind of evidence persuasive, relevant, or sufficient?**

Source: Chapter 7, §7.3

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By the end of class, you can

- distinguish ontology, epistemology, methodology, and axiology;
- compare how paradigms shape questions and evidence; and
- select a defensible combination of approaches for a contested decision.

Source: Chapter 7, §§7.3–7.3.3

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A paradigm is a working worldview

It shapes what a researcher takes to be real, how knowledge can be developed, which methods fit, and what purposes should guide inquiry.

Source: Chapter 7, §7.3.2

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Can one discipline use different paradigms?

Yes. A discipline is a field of inquiry; a paradigm is a set of assumptions that guides inquiry.

Two sociologists studying mountain lion management might ask:

- How do community norms relate to support for a proposed action?
- What meanings do residents attach to mountain lions?

Both work within sociology. They may begin from different paradigms.

Source: Chapter 7, §§7.2.1.2, 7.3.1-7.3.2; mountain lion example adapted for class.

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Four questions sit underneath a research design

Foundation	Plain-language question
Ontology	What is real here?
Epistemology	What can we know, and how?
Methodology	What process can produce credible knowledge?
Axiology	What values and responsibilities guide the inquiry?

Source: Chapter 7, §7.3.1

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Good research names its values and responsibilities

Researchers and managers make choices about:

- which problems are worth studying;
- whose experiences and knowledge are included;
- what harms to avoid; and
- how findings should be used.

Naming those choices makes research responsibilities more visible.

Source: Chapter 7, §7.3.1.

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The same mountain lion issue can generate different rigorous questions

Starting point	A question it makes possible
Measure a pattern	What share of residents supports a proposed action?
Interpret meaning	What do mountain lions mean to different communities?
Examine power	Whose voices are missing from management decisions?
Solve a problem	What evidence mix supports a workable decision?

Source: Chapter 7, §7.3.2; mountain lion example adapted for class.

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5 major research traditions relevant to HD

Paradigm	Starting point
Positivism	Measure a reality that exists independently.
Post-positivism	Measure carefully while accounting for uncertainty.
Interpretivism	Understand the meanings people make.
Critical paradigm	Examine how power shapes knowledge and decisions.
Pragmatism	Use approaches that fit the decision at hand.

They are **not the only paradigms**, and researchers still use them. Together they show how assumptions about measurement, uncertainty, meaning, power, and action shape research questions and evidence.

Source: Chapter 7, §§7.3.1–7.3.5.

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Positivism: measure a reality that exists independently

Where we start: the most measurement-focused tradition. Each paradigm that follows keeps or questions part of this view.

Reality exists independently of what people believe about it. Careful observation and experiments can reveal measurable patterns and effects.

A researcher could test whether a specific message changes support for a proposed mountain lion action.

Therefore: define the outcome, compare responses systematically, and measure the effect.

Source: Chapter 7, §7.3.2.1, mountain lion example adapted for class.

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Post-positivism: measure carefully and stay revisable

From positivism: keeps the belief that reality exists and the commitment to systematic evidence. Questions whether we can ever measure it exactly.

We can only approximate reality. Uncertainty, bias, and error shape every observation, so conclusions stay open to revision.

A survey can estimate support for a mountain lion action, but who responds and how the question is asked can affect the estimate.

Therefore: report uncertainty, check bias and alternative explanations, and revise conclusions when new evidence arrives.

Source: Chapter 7, §7.3.2.2, mountain lion example adapted for class.

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Interpretivism: examine how people make meaning

From post-positivism: a sharper break. There is no single reality to approximate; the same issue can hold multiple realities for different people.

People make meaning through experience, culture, and context. The same wildlife issue can mean different things to different people.

A researcher could ask residents what mountain lions and their management mean to them.

Therefore: gather open-ended accounts and interpret differences in context rather than force them into one average.

Source: Chapter 7, §7.3.2.3, mountain lion example adapted for class.

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Interpretive evidence is systematic, not "just stories"

Interviews, focus groups, participant observation, and narrative analysis can be designed carefully to examine how people construct meaning.

For mountain lion management, an interpretive study might ask how ranchers, Tribal members, agency staff, and urban residents each describe mountain lions and their management.

The goal is to understand context and differences in meaning—not to force them into a single account.

Source: Chapter 7.97.3.2.3 mountain lion example adapted for class.

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Critical paradigm: examine how power shapes knowledge

From interpretivism: keeps the idea that reality is socially constructed. Adds that power and privilege shape whose construction counts.

History and power shape whose knowledge is heard and who bears the consequences of a decision.

A researcher could ask whose concerns shaped a mountain lion management proposal and whose concerns were left out.

Therefore: examine exclusion and inequity, and ask whether the research reinforces or challenges them.

Source: Chapter 7.97.3.2.4 mountain lion example adapted for class.

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Pragmatism: choose evidence that serves the decision

From the other four: does not start from one objective truth or one constructed reality. It asks which knowledge is useful for this problem.

Knowledge is a tool for solving problems. The question, context, and constraints guide the choice of methods.

For a mountain lion decision, a team could combine a survey of support with interviews about reasons and meetings to test options.

Therefore: choose a credible evidence mix that fits the decision, budget, and timeline.

Source: Chapter 7.97.3.2.5 mountain lion example adapted for class.

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Indigenous knowledge systems are not an add-on data source

From the five paradigms: those are Western traditions. Indigenous knowledge systems answer the same four questions from their own foundations.

They are complete, place-based systems of knowledge, practice, relationship, responsibility, and governance.

Source: Chapter 7, §7.3.3

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Two-Eyed Seeing: work from the strengths of both eyes

Etuptumuk, often translated as *Two-Eyed Seeing*, is a Mi'kmaw concept that recognizes the value of Indigenous and Western knowledge systems while seeking respectful, meaningful partnership.

It is not a shortcut for extracting local observations into a Western project. It requires attention to relationships, sovereignty, and who guides the work.

Source: Chapter 7, §7.3.1.1

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Case diagnosis: choose evidence without choosing sides

Scenario: A state agency must revise a drought-season fishing closure. It has creel data, temperature records, and public comments, but hears that its current process misses subsistence and Tribal fishing perspectives.

In groups of three:

1. Name one question the agency must answer.
2. Name two kinds of evidence or knowledge that fit it.
3. State one process condition required for that evidence to be used responsibly.

Source: Chapter 7, §67.3-7.3.3

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Friday

Read the **Week 06 Case Brief: Four Proposals on One Desk – The Thornapple Lake Decision** before class. All case facts are in it; bring the concepts from Chapter 7 and this week's class work.

Four rigorous proposals, one \$180,000 budget, an eight-month deadline, and a late letter from a Tribal fisheries program setting a condition about who owns the knowledge.

Source: Chapter 7, §§7.3.1–7.3.3.

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Close: rigor is a match among question, evidence, and responsibility

Paradigms make our assumptions inspectable. That is useful because management decisions need evidence that is credible **and** a process that is appropriate to the people, places, and consequences involved.

Next week: use psychology to move from "people should care more" to a specific, observable behavior and a plausible strategy.

Source: Chapter 7, §§7.3–7.4.

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